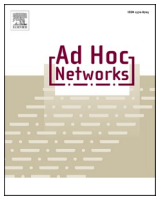
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A dynamic pricing scheme for secure offloading and resource allocation based on the internet of vehicles

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ABSTRACT

With the development of sixth-generation (6G) wireless communication technology, billions of vehicles will access the network in the future, and the number of vehicle applications and user data will also increase dramatically. Traditional cloud computing faces serious problems of latency and energy consumption in handling massive data. Mobile edge computing (MEC) has emerged to dramatically improve computing efficiency and reduce energy consumption by placing computing servers at network edge locations close to vehicles. However, the service range of MEC servers is limited and cannot fully satisfy user requirements. In addition, the task offloading process has security risks. The current research focuses on how to reduce the energy consumption and latency overhead of task offloading and neglects the economic cost or data transmission security of vehicles. To solve the above problems, we propose a cooperative security offloading (CSO) scheme for auxiliary vehicles and MEC servers. Firstly, we propose a dynamic pricing mechanism for computing resources by considering the credibility of MEC servers and auxiliary vehicles, the urgency of the task, and the number of users competing for auxiliary vehicles. Secondly, to prevent malicious MEC servers and eavesdroppers from attacking, we employ homomorphic encryption to protect user privacy. Meanwhile, efficient and secure computing services are achieved by optimizing user selection decisions, offloading decisions, and resource allocation decisions. Finally, the optimal decisions are obtained by the dueling DQN-based resource allocation and pricing strategy (DDRP) and the cost-minimizing security offloading (CMSO) algorithm, which minimizes the economic cost of users while maximizing security. Simulation results show that, compared with some existing schemes, the CSO scheme effectively reduces the economic cost of users while ensuring the security of data transmission.

1. Introduction

The Internet of Vehicles (IoV) is an emerging industrial form with deep integration of automobiles, electronics, information communication, road transportation, and other industries. It concentrates on the application of frontier technologies such as 5G, AI, big data, cloud computing, and blockchain to realize information interaction and data sharing among vehicles and between vehicles and road infrastructures [1,2]. However, due to the limited computing power of vehicles, it is challenging to meet the demands of delay-sensitive and computation-intensive tasks.

Mobile edge computing (MEC), as a new computing paradigm, effectively shortens the data transmission distance by deploying servers at roadside units (RSUs) or base stations (BSs) near the city to provide users with low-latency and high-reliability services. MEC effectively solves the problem of a mismatch between the computation and communication capabilities of vehicles and the task loads [3,4]. However, due to the high cost of infrastructure installation, RSUs are unable to reach the entire city [5]. Meanwhile, MEC servers have less computational power than central cloud servers [6]. In the future, as billions of vehicles are connected to the network, it will become crucial to fully utilize the idle resources of vehicles. Optimizing the energy efficiency of the system is also an important part of the communication system [7]. As a result, future research will focus on how to properly deploy computing resources to improve resource utilization.

In addition, the openness of the wireless offloading channel and the high mobility of vehicles may have some impact on the safety and stability of the offloading process. Eavesdroppers can easily access the task data offloaded to MEC servers from legitimate vehicles, and some honest and curious MEC servers will snoop on the private information of users during task offloading [8,9]. Therefore, another key challenge is how to protect user privacy while efficiently handling tasks in IoV.

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To address the above problems, we study the partial offloading model. Each user can process computational tasks through three execution modes: local computation, MEC offloading, and auxiliary vehicle offloading. Specifically, when users are within the coverage area of the MEC server, the computational tasks can be directly processed on the server. When users are outside the coverage area of the server, the computational tasks that need to be offloaded can be handled with the auxiliary vehicles. In addition, we homomorphically encrypt offloading data of users before offloading to ensure the security of user privacy.

Since the service price of MEC servers and auxiliary vehicles will affect the offloading decision of users, it is important to develop a reasonable pricing strategy [10]. We propose a dynamic pricing mechanism for computing resources by considering the credibility of MEC servers and auxiliary vehicles, the urgency of the task, and the number of users competing for auxiliary vehicles. Meanwhile, we consider the joint optimization of offloading decisions, user selection, and computational resource allocation to minimize the economic cost of users. The main contributions of this paper are summarized as follows.

1) In this paper, a Cooperative security offloading (CSO) scheme for auxiliary vehicles and MEC servers is proposed for the first time by comprehensively considering resource allocation, resource pricing, and security offloading issues in IoV. Outside the coverage of MEC servers, we utilize the free resources of auxiliary vehicles to provide offloading services for users to improve resource utilization and expand the communication range. Meanwhile, in task offloading, we utilize homomorphic encryption to protect user privacy while realizing data processing. Through the CSO scheme, we can jointly optimize the user selection decision, offloading decision, and resource allocation decision, and minimize the economic cost of users.
2) For eavesdroppers and malicious MEC servers in the channel, we adopt homomorphic encryption for the first time, which can directly compute the encrypted data on MEC servers and auxiliary vehicles to ensure the security of the user offloading process.
3) To address the resource allocation and pricing problems in IoV, we propose a dynamic pricing mechanism for computing resources, which takes into account the credibility of MEC servers and auxiliary vehicles, the urgency of the task, the number of users competing for auxiliary vehicles, etc. Adopting the dueling DQN-based resource allocation and pricing strategy (DDRP) algorithm and the cost-minimizing security offloading (CMSO) algorithm proposed in this paper, users can respectively obtain the best offloading decision to minimize the economic cost when offloading on MEC servers and auxiliary vehicles respectively, which ensures the rationality of resource pricing.

The rest of this paper is organized as follows. In Section 2, we discuss and analyze the latest related work. In Section 3, we construct the CSO model of auxiliary vehicles and MEC servers and develop a reasonable dynamic pricing mechanism. In Section 4, we propose the DDRP algorithm and CMSO algorithm to minimize the economic cost of users. The theoretical analysis in Section 5 proves the homomorphism of the encryption algorithm. Simulation results and conclusions are given in Sections 6 and 7, respectively.

2. Related work

In this section, we explore some of the existing research on security offloading and resource pricing in IoV.

2.1. Security offloading

Mobile vehicles are faced with the conflict between the demands of low-latency and low-energy tasks and the growing number of vehicle applications and user data. To cope with this challenge, mobile vehicles will process more tasks through offloading. However, due to the high mobility of vehicles and the openness of wireless channels, more and more users are concerned about privacy security during offloading [11].

The authors in [8] investigated a multi-vehicle multi-task non-orthogonal multiple access (NOMA)-assisted MEC system with passive eavesdropping vehicles. By introducing artificial noise to disturb eavesdropping vehicles, the transmission security is enhanced and the total delay of completion of tasks for edge vehicles is reduced. In [12], a joint security offloading and resource allocation scheme based on deep reinforcement learning is designed to reduce the system processing delay by utilizing physical layer security technology and spectrum sharing architecture. However, none of the works [8,12] considered malicious MEC servers. The authors of [13] proposed a blockchain-based security computational offload scheduling scheme, considering that malicious service providers may lead to significant security issues during vehicle offloading. The authors of [14] designed a distributed hierarchical software-defined wireless vehicular network framework to build a security architecture that reduces long-term latency, energy consumption, and traffic cost for all vehicles by jointly optimizing offloading decisions, consensus mechanism decisions, computing resource allocation, and channel bandwidth. The authors of [15] proposed an intelligent task cancellation algorithm based on online learning, which learns the optimal policy to balance task cancellation latency, queuing latency, and handover cost. However, it neglected the energy consumption of users. To achieve the goal of ensuring privacy security, reducing driver offloading time, and lowering energy consumption while maintaining edge load balancing, the authors of [16] designed a security service offloading method, which ensures data security and smooth service operation by using a locally sensitive hash algorithm for practical and privacy-aware service selection. However, the authors neglected the economic cost of users in [16].

All of the above studies have considered the privacy needs of users and have used different methods to protect user privacy to a certain extent. However, none of the above studies considered the economic cost of users or the resource pricing of MEC servers as compared to our work.

2.2. Resource pricing

When choosing to offload tasks to MEC servers, users must consider two important factors: resource allocation and resource pricing. With the rapid development of mobile Internet and the increasing user demands, users need more flexible and efficient computing methods, and MEC servers have limited resources compared to cloud servers. As a result, the resource allocation and pricing of MEC servers will also be limited and directly affect the offloading decisions of users.

Pricing strategies for vehicle-assisted MEC systems are investigated in [17]. Vehicle arrival and departure rates can be controlled by dynamically adjusting prices based on demand, thereby reducing the cost of the MEC system while meeting specific quality of service requirements. In the context of multi-user scenarios in intelligent IoV, the authors of [18] considered the budget constraints of the users and incorporated them into the design of the IoV system based on MEC. The design employs a deep Q-network (DQN) for offloading decisions and the computational power of the server is distributed among multiple users by utilizing the Lagrange multiplier method. The authors of [19] considered the partial offloading cost and the pricing of MEC servers. They aimed to determine the optimal vehicle offloading strategy by setting the service price and modeling the revenue. Additionally, they proposed resource allocation and pricing mechanisms for MEC servers to meet the computational resource constraints. The authors of [20] proposed a scheme to decentralize the computational tasks of industrial vehicles to multiple intelligent devices, including unmanned aerial vehicles (UAVs) and fixed-location MEC servers. This scheme reduces energy consumption, execution time, and the rental price of smart devices. In [21], the authors explored a joint optimization framework for

vehicular edge computing with multiple fog nodes, multiple data service agents, and multiple vehicle scenarios. Simulations are conducted by using Stackelberg games in a dynamic environment to solve the problem of fog node pricing and data service agent resource purchase. The authors of [22] investigated the trade-off between task processing latency and user cost and suggested a distributed computing offloading scheme for typical vehicle edge computing scenarios. A multi-leader and single-follower stack game model is designed to fully exploit the processing power of edge servers. An idle vehicle-assisted VEC server is suggested in [23] to compute offloading tasks, and a more flexible dynamic offloading scheme is suggested based on a thorough analysis of pricing and selection strategies. Through appropriate pricing, every VEC server is permitted to increase the system utility while optimizing the utility of idle vehicles. The authors of [24] proposed a dynamic pricing algorithm based on policy gradient (PG) for MEC systems. The algorithm can help access points (APs) optimize their revenues without the need to obtain private information about users or knowledge about system dynamics. In[25], the authors considered multiple MEC servers and multiple end-user scenarios to jointly optimize the perceived satisfaction of end-users and the price of MEC servers. The authors of [26] modeled D2D-assisted collaborative computational offloading for a single MEC server with multiple users and proposed a social relationship-based computational resource pricing method by jointing computational offloading and resource allocation schemes. In [27], the authors argued that intelligent transportation system (ITS) operator costs are affected by latency and limited computing resources. They considered queuing delays to reduce the ITS operator cost by optimizing the total energy consumption of servers subject to latency constraints.

The above research works considered the resource allocation and resource pricing of MEC servers from different perspectives. Especially in [19,22,23], the works fully considered the balance between delay and energy consumption and economic costs. However, compared to our work, the above studies did not consider privacy preservation in the offloading process and only the studies in [17,20,23,26] focused on idle resources in the system.

Mobile vehicles play a key role in ITS and need to process tasks in real-time to ensure efficiency and immediacy. However, the high cost of BS construction and the inability to achieve full coverage limit the service range of users. To address the above problem, we propose the CSO scheme between auxiliary vehicles and MEC servers, which provides task offloading services in areas without BS coverage to solve the latency problem, and users are required to pay the corresponding service fees. Homomorphic encryption is employed to protect user privacy from eavesdroppers and malicious MEC servers. In this paper, a dynamic pricing mechanism for computing resources is proposed by considering the credibility of MEC servers and auxiliary vehicles, the urgency of the task, and the number of users competing for auxiliary vehicles. Meanwhile, the offloading decision, user selection, and computational resource allocation are jointly optimized to minimize the economic cost to users.

3. System model and problem formulation

3.1. System model

In this system, there are M users, as shown by the red vehicles in Fig. 1, denoted by $\mathcal{M} = \{1,2,3,\ldots,M\}$; there are N auxiliary vehicles, i.e., the blue vehicles in the above figure, denoted by $\mathcal{N} = \{1,2,3,\ldots,N\}$; and there are J RSUs, each of which is equipped with a MEC server, denoted by $\mathcal{J} = \{1,2,3,\ldots,J\}$. In the MEC server coverage area represented by the blue area in the figure, the server can serve multiple users at the same time. These users will incur a corresponding offloading service charge by utilizing server resources. In addition, auxiliary vehicles can extend the service to users outside the coverage area of the server, which also requires the payment of offloading service fees. If the users wish, the task can also be processed locally without paying any offloading fees.

The task U_m characteristics of the user m can be represented by (s_m, c_m,t_m), where s_m is the data size of the task in bits; t_m is the maximum tolerable delay in seconds; and c_m is the CPU cycles required to process the unit task in cycles/bit. In this paper, we adopt a partial offloading protocol and define the offloading variables of the user m as $\alpha_m = \left\{\alpha_m^l, \alpha_m^j, \alpha_m^n\right\}$, which denote the offloading ratio of the local processing, offloading to the MEC server, and offloading to auxiliary vehicles, respectively. Therefore, $\boldsymbol{\alpha} = \{\alpha_1,\cdots, \alpha_m,\cdots, \alpha_M\}$ denotes the offloading decision vector.

3.2. Local computing model

The time consumed by the user m while locally processing task U_m can be expressed as

$$t_m^l = \frac{\alpha_m^l s_m c_m}{f_m}, \tag{1}$$

where f_m (in cycles per second) is the local CPU frequency of the user m.



Fig. 1. System model diagram.

The local computing energy consumption of the user m can be expressed as

$$E_m^l = \kappa_i (f_m)^2 \alpha_m^l s_m c_m, \tag{2}$$

where $\kappa_i (f_m)^2$ is the local CPU energy consumption per cycle, $\kappa_i > 0$ and is a coefficient associated with the architecture of the chip [28].

3.3. Privacy protection model

While MEC technology has many benefits in terms of reducing energy consumption and delay, information security is also a concern when it comes to task offloading [29]. Due to the broadcast nature of cellular wireless communication channels, it is simple for eavesdroppers to collect task data which is offloaded by authorized vehicles to MEC servers [8]. Therefore, the presence of malicious eavesdroppers in MEC systems must be considered, and appropriate measures must be taken to ensure the security of offloading information. In addition, the MEC server, as a public ecosystem, is managed by different service providers [30]. The authors of [31] mentioned that when edge servers are compromised or untrusted, the security of user privacy can be violated, and the consequences of this privacy violation will go beyond information leakage. In this paper, to safeguard the privacy of offloading information of the user m, we utilize homomorphic encryption to encrypt the task before the user offloads it.

Homomorphic encryption is seen as a prospective technology to tackle the above challenges because it allows computations to be performed on encrypted data [32]. In particular, homomorphic encryption allows clients to perform computations securely on an untrusted server, which processes the ciphertext without decrypting it and returns the encrypted result to the clients. This means that even if the data is eavesdropped on during transmission, it is impossible to access the plaintext information. Meanwhile, for an untrusted server, even if it can process the ciphertext and return the result, it will not be able to access or tamper with the original data because it will not be able to decrypt that data. Therefore, homomorphic encryption can solve the problems of channel eavesdropping and malicious servers while protecting data confidentiality. The specific encryption scheme is as follows:

Step 1: Generate the key.

Two independent large prime numbers o and q are chosen at random and satisfied $gcd(oq, (o-1)(q-1)) = 1$, such that o and q are of equal length. $gcd(o,q)$ denotes the greatest common factor of o and q. Compute $h = oq$; $l = lcm(o-1, q-1)$, $lcm(o,q)$ denotes the least common multiple of o and q. Randomly choose integer i, $i \in \mathbb{Z}_{n^2}^*$, and satisfy $\mu = \left(L\left(i^l \bmod h^2\right)\right)^{-1} (\bmod h)$, where $L(x) = \frac{x-1}{h}$. At this point, the public key is (h, i), and the private key is(l, μ).

Step 2: Encryption Algorithm.

For the plaintext X, $X \in \mathbb{Z}_n$, a random number $\eta < h$ is chosen, and the encryption process is $Y = i^X \eta^h \left(\bmod h^2\right)$.

Step 3: Decryption Algorithm.

The decryption process of ciphertext Y is:

$$X = L\left(Y^l \left(\bmod h^2\right)\right) * \mu (\bmod h). \tag{3}$$

For the user m, the encryption time and the energy consumption used to protect the data can be respectively expressed as

$$t_m^{en} = \frac{s_m c_m^{en}}{f_m}, \tag{4}$$

$$E_m^{en} = \kappa_i (f_m)^2 s_m c_m^{en}, \tag{5}$$

where c_m^{en} denotes the number of CPU cycles required for the encryption task.

3.4. Task offloading models

By offloading tasks when the computational resources of the vehicle are insufficient to handle a large number of tasks in a timely way, the vehicle can utilize additional computing power and resources in the external environment to ensure efficient and timely execution of tasks. Fig. 2 illustrates the details of user task offloading.

(1) Offloading tasks to the MEC server

As shown in Fig. 2, when a vehicle passes through the area covered by the MEC server, the distance traveled by the vehicle can be expressed as

$$x = 2\sqrt{y^2 - z^2}, \tag{6}$$

where z denotes the distance from the MEC server to the curb and y denotes the radius of the coverage of the MEC server.

The residence time of the user m within the MEC service area is critical in determining the resource allocation policy of the MEC server, which influences how the MEC server allocates its resources to efficiently handle offloading tasks. The residence time of the user m can be expressed as

$$t_m^w = \frac{x - v_m t_q}{v_m}, \tag{7}$$

where v_m (in m/s) is the average speed of the user m. t_q denotes the travel time within the coverage area of the MEC server before the user m offloads the task. $v_m t_q$ denotes the distance traveled within the coverage area of the MEC server before the user m offloads the task.

In this paper, OFDMA technology is used in the communication link, and we assume that there is no signal interference between the vehicles. Therefore, the transmission rate is primarily impacted by the variation of the distance between the user m and the MEC server. The distance between the user m and the MEC server can be expressed as

$$u_m(t) = \sqrt{z^2 + \left(\frac{x}{2} - v_m t_m^w\right)^2}. \tag{8}$$

Therefore, the uplink instantaneous transmission rate of the user m can be written as

$$r_m(t) = B\log_2\left(1 + \frac{p_m {u_m}^{-\theta}(t)\delta_0}{W_0}\right), \tag{9}$$

where B is the bandwidth of each OFDMA subchannel, p_m is the transmit power of the user m, and δ_0 is the channel gain over the reference distance [33]. In addition, θ is the path loss exponent, and W_0 is the background noise power spectral density of each user channel.

The task processing of users on the MEC server is divided into three phases: task uploading, task processing, and result backhaul. Because the size of the result data is typically relatively tiny and the downlink transmission rate is substantially larger than the uplink transmission rate, Similar to other studies [34,35], we neglect the time required for the resulting backhaul.

When the user m offloads the task U_m to the MEC server j, the delay and energy consumption of the task uploading phase can be respectively expressed as

$$t_{m,j}^{up} = \frac{\alpha_m^j s_m}{r_m(t)}, \tag{10}$$

$$E_{m,j}^{up} = \frac{\alpha_m^j s_m}{r_m(t)} p_m. \tag{11}$$

The latency of the task processing phase can be expressed as

$$t_{m,j}^{exe} = \frac{\alpha_m^j s_m c_m}{\beta_m f_j}, \tag{12}$$

where f_j (in cycles per second) denotes the computational capacity of the

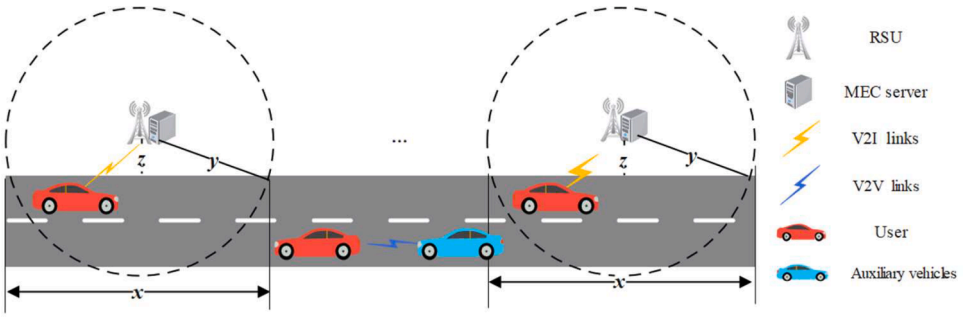


Fig. 2. Detailed diagram of user task offloading.

MEC server, β_m denotes the percentage of computing resources allocated to the user m, and $\boldsymbol{\beta} = \{\beta_1, \ldots, \beta_m, \ldots, \beta_M\}$ denotes the resource allocation vector.

In addition, when the MEC server processes the uploaded tasks, the idle power of the user is denoted by p_w. Therefore, the waiting energy consumption of the user m can be expressed as

$$E_m^w = p_w t_{m,j}^{exe} = \frac{p_w \alpha_m^j s_m c_m}{\beta_m f_j}. \tag{13}$$

The total energy consumption for offloading tasks to the MEC server can be expressed as

$$E_m^j = E_m^{en} + E_{m,j}^{up} + E_m^w = \kappa_i (f_m)^2 \alpha_m^j s_m c_m^{en} + \frac{\alpha_m^j s_m}{r_m(t)} p_m + \frac{p_w \alpha_m^j s_m c_m}{\beta_m f_j}. \tag{14}$$

During the path traveled by the user in Fig. 2, certain areas are located within the communication range of the MEC server, while other areas are out of their communication range. However, auxiliary vehicles can be utilized to offload, to ensure timely processing of the task. In this paper, the distance between the user and the MEC server is used as the basis for offloading selection. In addition, the user can choose from a variety of auxiliary vehicles and prioritize the auxiliary vehicles with abundant resources and lower costs, and the service of one auxiliary vehicle is limited to one user.

(2) Offloading tasks to auxiliary vehicles

The delay in offloading tasks to the auxiliary vehicle n consists of three parts: uploading delay, processing task delay, and data backhaul delay. The uploading delay and energy consumption can be respectively expressed as

$$t_{m,n}^{up} = \frac{\alpha_m^n s_m}{r_{m,n}^{up}(t)}, \tag{15}$$

$$E_{m,n}^{up} = \frac{\alpha_m^n s_m}{r_{m,n}^{up}(t)} p_m, \tag{16}$$

where $r_{m,n}^{up}(t)$ is the uplink transmission rate from the user m to the vehicle n, which can be expressed as

$$r_{m,n}^{up}(t) = B_{V2V} \log_2 \left(1 + \frac{p_m d_{m,n}{}^{-\theta}(t) \delta_0}{W_0}\right), \tag{17}$$

where B_{V2V} is the bandwidth of the V2V communication, $d_{m,n}(t)$ is the distance between the user m and the vehicle n.

Similarly, f_m^n is denoted as the computational power allocated by the auxiliary vehicle n to the task U_m. The computational latency of the task is therefore the same as that of the task on the auxiliary vehicle. Therefore, the computational latency of executing a task U_m on the vehicle n can be expressed as

$$t_{m,n}^{exe} = \frac{\alpha_m^n s_m c_m}{f_m^n}. \tag{18}$$

In addition, the idle power of the user is denoted by p_w when the auxiliary vehicle handles the uploaded task. Therefore, the energy consumption of the user m can be expressed as

$$E_m^w = p_w t_{m,n}^{exe} = \frac{p_w \alpha_m^n s_m c_m}{f_m^n}. \tag{19}$$

The total energy consumption of the offloading task to the auxiliary vehicle can be expressed as

$$E_m^n = E_m^{en} + E_{m,n}^{up} + E_m^w = \kappa_i (f_m)^2 \alpha_n s_m c_m^{en} + \frac{\alpha_n s_m}{r_{m,n}^{up}(t)} p_m + \frac{p_w \alpha_m^n s_m c_m}{f_m^n}. \tag{20}$$

The total time constraint to complete the task can be expressed as

$$\begin{aligned} t_{all} &= \max\left\{t_m^l, t_{m,j}^{up} + t_{m,j}^{exe} + t_m^{en}, t_{m,n}^{up} + t_{m,n}^{exe} + t_m^{en}\right\} \\ &= \max\left\{\frac{\alpha_m^l s_m c_m}{f_m}, \frac{\alpha_m^j s_m}{r_m(t)} + \frac{\alpha_m^j s_m c_m}{\beta_m f_j} + \frac{\alpha_m^j s_m c_m^{en}}{f_m}, \frac{\alpha_m^n s_m}{r_{m,n}^{up}(t)} + \frac{\alpha_m^n s_m c_m}{f_m^n} + \frac{\alpha_m^n s_m c_m^{en}}{f_m}\right\} \\ &\le t_m \end{aligned} \tag{21}$$

3.5. Economic cost models

In this paper, we define a user-selected variable $\varphi_m = \left\{\varphi_m^j, \varphi_m^n\right\}_{m \in M}$. $\varphi_m^j = 1$ if the user m offloads the task to the MEC server j, $\varphi_m^j = 0$ otherwise. $\varphi_m^n = 1$ if the user m offloads the task to the auxiliary vehicle n, $\varphi_m^n = 0$ otherwise. $\boldsymbol{\varphi} = \{\varphi_1, \ldots, \varphi_m, \ldots, \varphi_M\}$ is a user-selected offloading vector, and users cannot offload tasks to the MEC server at the same time as the auxiliary vehicle.

The price of the offloading service for each cycle of the MEC server is denoted as $g_j = b_j \cdot \exp(\omega_m) \cdot g_{j0}$, where $b_j \in (0, 1)$ is the credibility of the MEC server, $\omega_m \in (0, 1)$ is the urgency of the task U_m, and g_{j0} is (in cents/cycle) the price of the offloading service, which is defined as the payment to the MEC server per CPU cycle. The unit price of the offloading service for the auxiliary vehicle is denoted as $g_n = b_n \cdot \ln(1 + \varepsilon_n) \cdot \omega_m \cdot g_{n0}$, where $b_n \in (0, 1)$ is the credibility of the auxiliary vehicle, ε_n is the number of users requesting offloading to the auxiliary vehicle n at the same time; the larger ε_n, the higher the offloading price of the auxiliary vehicle; and g_{n0} is (in cents/cycle) the price of the offloading service,

which is defined as the payment to the auxiliary vehicle per CPU cycle.

Therefore, the economic cost of the user m can be defined as

$$\begin{aligned} C_m &= \lambda_m^l \cdot E_m^l + \varphi_m^j \left(\lambda_m^j E_m^j + g_j \alpha_m^j s_m c_m\right) + \varphi_m^n \left(\lambda_m^n E_m^n + g_n \alpha_m^n s_m c_m\right) \\ &= \lambda_m^l \kappa_i (f_m)^2 \alpha_m^l s_m c_m \\ &\quad + \varphi_m^j \left(\lambda_m^j \left(\kappa_i (f_m)^2 \alpha_m^j s_m c_m^{en} + \frac{\alpha_m^j s_m}{r_m(t)} p_m + \frac{p_w \alpha_m^j s_m c_m}{\beta_m f_j}\right) + b_j \exp(\omega_m) g_{j0} \alpha_m^j s_m c_m\right) \\ &\quad + \varphi_m^n \left(\lambda_m^n \left(\kappa_i (f_m)^2 \alpha_m^n s_m c_m^{en} + \frac{\alpha_m^n s_m}{r_{m,n}^{up}(t)} p_m + \frac{p_w \alpha_m^n s_m c_m}{f_m^n}\right) + b_n \ln(1+\mu_n) \omega_m g_{n0} \alpha_m^n s_m c_m\right) \end{aligned} \tag{22}$$

where the energy consumption of the vehicle is converted to a monetary expense by using conversion factors λ_m^l, λ_m^j, and λ_m^n (in cents/joule), which represent the price per joule of energy. For energy-sensitive vehicles, λ_m^l and λ_m^n will take a higher value compared to other continuously powered vehicles.

We reduce the economic cost for all users by optimizing the offloading decision vector $\boldsymbol{\alpha}$, the resource allocation vector $\boldsymbol{\beta}$, and the user-selected offloading vector $\boldsymbol{\varphi}$. Thus, the optimization problem can be formulated as

$$\mathrm{Q1}: \min_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\varphi}} \sum_{m=1}^{M} C_m \tag{23a}$$

$$\text{s.t.} \alpha_m^l + \alpha_m^n = 1, \tag{23b}$$

$$\alpha_m^l + \alpha_m^j = 1, \tag{23c}$$

$$\alpha_m^l \in (0,1), \alpha_m^n \in (0,1), \alpha_m^j \in (0,1), \forall m \in \boldsymbol{M}, \tag{23d}$$

$$\max\left\{t_{m,l}, t_{m,j}^{up} + t_{m,j}^{exe} + t_m^{en}, t_{m,n}^{up} + t_{m,n}^{exe} + t_m^{en}\right\} \le t_m, \tag{23e}$$

$$\varphi_m^j = \{0,1\}, \varphi_m^n = \{0,1\}, \forall m \in \boldsymbol{M}, \tag{23f}$$

$$\varphi_m^j + \varphi_m^n = 1, \tag{23 g}$$

$$0 \le f_m \le f_m^{\max}, \tag{23h}$$

$$0 \le f_m^n \le f^n, \tag{23i}$$

$$\sum_{m=1}^{M} \beta_m \le 1. \tag{23j}$$

(23-b) and (23-c) ensure that a task cannot be offloaded to both the MEC server and the auxiliary vehicle. (23-d) constrains the range of task offloading ratios. (23-e) is the latency constraint that ensures that the maximum allowable delay is not exceeded by the total processing time. (23-f) and (23-g) are the user-selected constraints that ensure that the user selection of task offloading is limited to either the MEC server or the auxiliary vehicle. (23-h) is the constraint on the computational power of the user. (23-i) makes sure that the computing resources allotted by the vehicle n to the user m cannot be more than the total resources of the vehicle n. (23-j) is the constraint on the available computational resources, i.e., the sum of the computational resources allocated to each user cannot exceed the total computational resources of the MEC server.

4. Problem analysis and resolution

In this section, we will first outline the user-selected offloading target algorithm to help the user select the offloading target. Then we have a categorical discussion to analyze the decision problem of the user offloading the task to the MEC server and the auxiliary vehicle.

4.1. Algorithm for user selection of offloading targets

For the user m, we first determine whether it is within the coverage range of the MEC server based on $u_m(t)$. Given that the maximum coverage range of the MEC server is $d_{\max}$, if $d_{m,j} \le d_{\max}$, the user task is offloaded to the MEC server; otherwise, the user will choose among the

Algorithm 1
Algorithm for user selection of offloading targets

1. **Input:** The maximum coverage range of the MEC server $d_{\max}$.
2. Initialization: user set $\mathscr{M} = \{1,2,3,\ldots,M\}$, auxiliary vehicle set $\mathscr{N} = \{1,2,3,\ldots,N\}$, MEC server set $\mathscr{J} = \{1,2,3,\ldots,J\}$, user-selected offloading variable $\varphi = \left\{\varphi_m^j, \varphi_m^n\right\}_{m\in M} = \{0,0\}$;
3. **for** $m = 1$ to M **do**
4. **for** $j = 1$ to J **do**
5. Calculate the distance $d_{m,j}$ between the user m and the MEC server j;
6. **if** $d_{m,j} \le d_{\max}$ **then**
7. $\varphi = \left\{\varphi_m^j, \varphi_m^n\right\}_{m\in M} = \{1,0\}$ and $n_m^* = 0$;
8. **else**
9. **for** $n = 1$ to N **do**
10. $\varphi = \left\{\varphi_m^j, \varphi_m^n\right\}_{m\in M} = \{0,1\}$;
11. Calculate the price of offloading services for each auxiliary vehicle $g_n = b_n \cdot \ln(1+\varepsilon_n) \cdot \omega_m \cdot g_{n0}$;
12. **end for**
13. Choose the vehicle with the lowest service price $n_m^* = \min_{n\in N}[g_n]$;
14. **end if**
15. **end for**
16. **end for**
17. **Output:** $\boldsymbol{\varphi} = \{\varphi_1,\ldots,\varphi_m,\ldots,\varphi_M\}$, $\mathbf{n}^* = \{n_1^*,\ldots,n_m^*,\ldots,n_M^*\}$.

nearby auxiliary vehicles, and the user will calculate the offloading service price of each auxiliary vehicle and prioritize the auxiliary vehicle with the lowest price. The process of selecting the offloading target by the user is described in detail in Algorithm 1.

Through Algorithm 1, we can determine the user selection of offloading targets. We will discuss the issues related to user offloading tasks to MEC servers in 4.2 and offloading to auxiliary vehicles in 4.3.

4.2. MEC server resource allocation and pricing

In this section, we discuss the problem of resource allocation and pricing for MEC servers, where the objective function can be transformed as

$$\begin{aligned} C_m^j &= \lambda_m^l E_m^l + \lambda_m^j E_m^j + g_j \alpha_m^j s_m c_m \\ &= \lambda_m^l \left(\kappa_i (f_m)^2 \alpha_m^l s_m c_m + \kappa_i (f_m)^2 \alpha_m^j s_m c_m^{en} + \frac{\alpha_m^j s_m}{r_m(t)} p_m \right) \\ &\quad + \lambda_m^j \frac{p_w \alpha_m^j s_m c_m}{\beta_m f_j} + b_j \exp(\omega_m) g_{j0} \alpha_m^j s_m c_m \end{aligned} \tag{24}$$

Thus, the problem Q1 can be transformed as

$$\mathrm{Q2}: \min_{\alpha,\beta} \sum_{m=1}^{M} C_m^j \tag{25a}$$

$$\text{s.t.} \alpha_m^l + \alpha_m^j = 1, \tag{25b}$$

$$\alpha_m^l \in (0,1), \alpha_m^j \in (0,1), \forall m \in M, \tag{25c}$$

$$\max\left\{ t_{m,l}, t_{m,j}^{up} + t_{m,j}^{exe} + t_m^{en} \right\} \le t_m, \tag{25d}$$

$$\sum_{m=1}^{M} \beta_m \le 1, \tag{25e}$$

$$0 \le f_m \le f_m^{\max}. \tag{25f}$$

(25-b) is the constraint between local computation and the offloading ratio of offloading to the MEC server. (25-c) constrains the range of the offloading ratio. (25-d) is the latency constraint, which ensures that the total processing time does not exceed the maximum tolerable latency. (25-e) is the constraint on the available computational resources, i.e., the sum of the computational resources allocated to each user must not exceed the total computational resources of the MEC server. (25-f) denotes the limitation on the computational power of the user.

To reduce the economic cost in a dynamic environment, the above optimization problem can be modeled by using a Markov decision process (MDP). In this process, state space, action space, and reward function are defined and then optimized by using the dueling DQN algorithm.

1) The state space is defined as $S = [s_1, \cdots, s_m, \cdots, s_M; t_1^w, \cdots, t_m^w, \cdots, t_M^w]$, denotes the user task size and the time spent in the coverage area of the MEC server j, respectively.
2) Action space: we reduce the economic cost of the user m by formulating strategies for computational offloading and resource allocation. In this paper, we adopt a partial offloading protocol that enables the user m to upload any percentage of tasks to the server and enables the server to assign any percentage of computing resources to the user m. The action space is defined as $a = [\alpha_1, \cdots, \alpha_m, \cdots, \alpha_M; \beta_1, \cdots, \beta_m, \cdots, \beta_M]$.
3) Reward function: the reward function R is used to measure the advantages and disadvantages of taking action a in state S. Moreover, the purpose of the essential algorithm of DRL is to maximize the reward by taking reasonable action in any state S. Therefore, the reward function can be expressed as $R(S,a) = -\sum_{m=1}^{M} C_m^j$.

The actions of an intelligent body must adhere to the limitations stated in (25). The reward function is given a penalty value if the constraints are broken. Define the state history as $H_t = \{S_1, S_2, S_3, \cdots, S_t\}$. If $pr(S_{t+1}|S_t) = pr(S_{t+1}|H_t)$, the state is a Markov chain. MDPs are frequently used as models for dynamic programming issues.

The reward is used to evaluate the performance of the action a_t in the state S_t to reach the next state S_{t+1}. There is a difference between the reward now and the reward later. Consequently, it is necessary to incorporate a discount factor ψ. The total benefits can be written as

$$R_t^{sum} = R_{t+1} + \psi R_{t+2} + \cdots + \psi^{T-t-1} R_T, \tag{26}$$

where $\psi = 1$ indicates that the future reward is equal to the current reward, and $\psi = 0$ indicates that only the current reward is of interest.

The dueling DQN algorithm defines the result of subtracting the state value function V from the action value function Q as the dominance function A, which reflects the correlation between the action value $Q(S, a)$ in state S and the selected action a. The dueling DQN algorithm is based on the following model:

$$A(S,a) = Q(S,a) - V(S). \tag{27}$$

The dominance function has two important properties. Firstly, the expectation of the dominance function concerning the distribution $a \sim \pi(s|a)$ is 0 for a given strategy π, i.e., $\mathbb{E}_{a\sim\pi(s|a)}[A(s,a)] = 0$. Secondly, for a deterministic strategy π, the dominance function is $A(s, a) = 0$ for a selected action $a' = \pi(s)$ in state s.

Thus, in dueling DQN algorithm, the Q-network is modeled as

$$Q(S,a;\vartheta) = V(S;\vartheta) + A(S,a;\vartheta), \tag{28}$$

where ϑ is a network parameter shared by the state value function and the dominance function. In the implementation, we replace the maximization operation with averaging, i.e.,

$$Q(s,a;\theta) = V(s;\theta) + A(s,a;\theta) - \frac{1}{|A|} \sum_{a'} A(s,a';\theta). \tag{29}$$

The DDRP algorithm in this paper is summarized in Algorithm 2.

4.3. Auxiliary vehicle resource pricing

In this section, we will discuss the problem of resource pricing of auxiliary vehicles, the objective function can be transformed as

$$\begin{aligned} C_m &= \lambda_m^l E_m^l + \varphi_m^n \left(\lambda_m^n E_m^n + g_n \alpha_m^n s_m c_m \right) \\ &= \lambda_m^l \left(\kappa_i (f_m)^2 \alpha_m^l s_m c_m + \kappa_i (f_m)^2 \alpha_m^n s_m c_m^{en} + \frac{\alpha_m^n s_m}{r_{m,n}^{up}(t)} p_m \right) \\ &\quad + \lambda_m^n \frac{p_w \alpha_m^n s_m c_m}{f_m^n} + b_n \ln(1+\varepsilon_n) \omega_m g_{n0} \alpha_m^n s_m c_m \end{aligned} \tag{30}$$

Thus, problem Q1 can be converted as

$$\mathrm{Q3}: \operatorname*{Min}_{\alpha} \sum_{m=1}^{M} C_m^n \tag{31a}$$

$$\text{s.t.} \alpha_m^l + \alpha_m^n = 1, \tag{31b}$$

$$\alpha_m^l \in (0,1), \alpha_m^n \in (0,1), \forall m \in M, \tag{31c}$$

$$\max\left\{ t_{m,l}, t_{m,n}^{up} + t_{m,n}^{exe} + t_m^{en} \right\} \le t_m, \tag{31d}$$

$$0 \le f_m \le f_m^{\max}, \tag{31e}$$

$$0 \le f_m^n \le f^n. \tag{31f}$$

Algorithm 2
Dueling DQN-based Resource Allocation and Pricing Strategy

Input: number of iterations N_t, time step T, discount factor γ, experience playback library D, frequency of network parameter updates ς.
Random initialization of neural networks $Q(S,a;\vartheta)$, and the parameters ϑ;
Initialize the target network Q' with the parameter $\vartheta' \leftarrow \vartheta$;
Initialization of the experience playback library D;
Initializing the probability of exploration ρ;
for episode = 1 to N_t **do**
  Initialize the user task size and stay time;
  Randomized acquisition of states S_t;
  for $t = 1$ to T **do**
    The random selection probability σ;
    if $\sigma \le \rho$ **then**
      Randomly select an action a_t;
    else
      Select an action $a_t = \frac{1}{|A|}\sum_{a'} Q(s,a';\vartheta)$;
    end if
    According to a_t, determine the offloading decision vector $\boldsymbol{\alpha}$ and the resource allocation vector $\boldsymbol{\beta}$;
    Execute the appropriate strategy, get the reward R_t, and observe the next state S_{t+1};
    Save conversion S_t, a_t, R_t, S_{t+1} to the experience playback library D;
    Samples are drawn from the experience playback library D to update the network parameters;
  end for
  end for
Output: $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, the minimum economic cost of the user m C_m^i.

(31-b) is the constraint between local computation and the offloading ratio of offloading to the auxiliary vehicle. (31-c) constrains the range of the offloading ratio. (31-d) is the latency limitation that assures total processing time does not exceed the maximum tolerated latency. (31-e) denotes the constraint on the computational power of the user. (31-f) is a guarantee that the computational resources allocated by the vehicle n to the user m cannot exceed all the resources of the vehicle n.

From (31-d) and (31-e), we can obtain $t_{m,l} \le t_m$, and then lead to:

$$\alpha_m^l \le \frac{t_m f_m}{s_m c_m} \underline{\underline{\Delta}} \left(\alpha_m^l\right)^{ub}. \tag{32}$$

When the user m selects to offload some of the tasks to the MEC server, the transmission time $t_{m,n}^{up}$ must be less than the deadline time of the task t_m, and therefore

$$\alpha_m^l \ge 1 - \frac{t_m R_{m,n}^{up}(t)}{s_m} \underline{\underline{\Delta}} \left(\alpha_m^l\right)^{lb}. \tag{33}$$

At this point, $\left(\alpha_m^l\right)^{\max} \underline{\underline{\Delta}} \min\left\{\left(\alpha_m^l\right)^u, 1\right\}$, $\left(\alpha_m^l\right)^{\min} \underline{\underline{\Delta}} \max\left\{0, \left(\alpha_m^l\right)^l\right\}$.

In (30), C_m^n monotonically increases with f_m. According to the constraint (31-d), we can obtain $t_{m,l} \le t_m$, hence $f_m \ge \alpha_m^l s_m c_m / t_m$ and the optimal local CPU frequency f_m^* can be expressed as

$$f_m^* = \frac{\alpha_m^l s_m c_m}{t_m}. \tag{34}$$

And we substitute $\alpha_m^l = 1 - \alpha_m^n$ into (30). At this point, (30) is transformed as

$$\begin{aligned} C_m^n &= \lambda_m^l E_m^l + \varphi_m^n\left(\lambda_m^n E_m^n + g_n \alpha_m^n s_m c_m\right) \\ &= \lambda_m^l \left(\kappa_i \frac{\left(\alpha_m^l s_m c_m\right)^3}{t_m^2} + \kappa_i \frac{\left(\alpha_m^l s_m c_m\right)^2}{t_m^2}\left(1-\alpha_m^l\right) s_m c_m^{en} + \frac{\left(1-\alpha_m^l\right) s_m}{r_{m,n}^{up}(t)} p_m\right) \\ &\quad + \lambda_n \frac{p_w\left(1-\alpha_m^l\right) s_m c_m}{f_m^n} + b_n \ln(1+\varepsilon_n)\omega_m g_{n0}\left(1-\alpha_m^l\right) s_m c_m. \end{aligned} \tag{35}$$

The first and second-order derivatives of C_m^n can be written as

$$\begin{aligned} \frac{\partial C_m^n}{\partial \alpha_m^l} &= \lambda_m^l \left(\kappa_i \frac{3\left(\alpha_m^l\right)^2 \left(s_m c_m\right)^3}{t_m^2} + \kappa_i \frac{\left(2\alpha_m^l - 3\left(\alpha_m^l\right)^2\right)\left(s_m\right)^3 \left(c_m\right)^2 c_m^{en}}{t_m^2} - \frac{s_m p_m}{r_{m,n}^{up}(t)}\right) \\ &\quad - \lambda_n \frac{p_w s_m c_m}{f_m^n} - b_n \ln(1+\varepsilon_n)\omega_m g_{n0} s_m c_m, \end{aligned} \tag{36}$$

$$\begin{aligned} \frac{\partial^2 C_m^n}{\partial\left(\alpha_m^l\right)^2} &= \lambda_m^l \left(\kappa_i \frac{6\left(\alpha_m^l\right)\left(s_m c_m\right)^3}{t_m^2} + \kappa_i \frac{\left(2-6\alpha_m^l\right)\left(s_m\right)^3 \left(c_m\right)^2 c_m^{en}}{t_m^2}\right) \\ &= \lambda_m^l \kappa_i \left(\frac{6\alpha_m^l \left(s_m\right)^3 \left(c_m\right)^2 \left(c_m - c_m^{en}\right)}{t_m^2} + \frac{2\left(s_m\right)^3 \left(c_m\right)^2 c_m^{en}}{t_m^2}\right) \ge 0. \end{aligned} \tag{37}$$

Therefore, (30) is a convex problem, and all constraints are convex, so this optimization problem can be solved directly with existing optimization tools, such as CVX in MATLAB [36].

The CMSO algorithm, as demonstrated in Algorithm 3, may find the optimal task offloading ratio that minimizes the economic cost of the user m.

5. Theoretical analysis

In this section, we prove the additive homomorphism of the CSO

Algorithm 3
Cost Minimizing Security Offloading Algorithm

Input: $f_m, s_m, c_m, c_m^{en}, p_m, r_{m,n}^{up}(t), p_w, f_m^n, b_n, \varepsilon_n, \omega_m, g_{n0}$.
According to Algorithm 1, determine the auxiliary vehicle n_m^*;
Calculate the range α_m^l from (32) and (33), i.e., $\left(\alpha_m^l\right)^{ub}$ and $\left(\alpha_m^l\right)^{lb}$;
Calculate the optimal local CPU frequency f_m^* from (34);
Use CVX to solve $\left(\alpha_m^l\right)^*$ and compute the optimal economic cost $\left(C_m^n\right)^*$;
Output: $\left(\alpha_m^l\right)^*$ and $\left(C_m^n\right)^*$.

scheme.

For plaintexts X_1,X_2, we randomly select η_1,η_2 and η_1,$\eta_2 < h$. After encryption using the public key (h, i), the ciphertext Y_1,Y_2 is obtained as

$$Y_1 = E(X_1;\eta_1) = g^{X_1}\eta_1^h (\bmod h^2), \tag{38}$$

$$Y_2 = E(X_2;\eta_2) = g^{X_2}\eta_2^h (\bmod h^2), \tag{39}$$

$$\begin{aligned} Y_1 \cdot Y_2 &= E(X_1;\eta_1)\cdot E(X_2;\eta_2) \\ &= \left(g^{X_1}\eta_1^h\right)\left(g^{X_2}\eta_2^h\right)(\bmod h^2) \\ &= g^{X_1+X_2}(\eta_1\eta_2)^h \bmod h^2 \\ &= E(X_1+X_2;\eta_1,\eta_2). \end{aligned} \tag{40}$$

Then we obtain $E(X_1; \eta_1) \cdot E(X_2; \eta_2) = E(X_1 + X_2; \eta_1,\eta_2)$ in (40), i.e., the result of multiplying the ciphertext is decrypted to get the result of adding the plaintext, and the encryption algorithm in this paper satisfies the additive homomorphism.

6. Simulation analysis

In this section, the performance of the CSO scheme is compared with that of the random offloading scheme, the AES scheme, and the cost optimization scheme in [19]. Next, we will introduce the settings of simulation parameters and three comparison schemes in detail.

Firstly, the number of vehicles is assumed to be from 1 to 4, the vehicle speed is assumed to be from 15 m/s to 25 m/s, and the radius of the service area of the MEC server is chosen as $L = 300$ m. The computational power of the MEC server is 1 GHz, the computational power of the vehicle is 0.5 GHz, the number of CPU cycles is 1000 cycles/bit, and the number of homomorphic encrypted CPU cycles is 800 cycles/bit. The transmitting power of the vehicle is 0.2 W, the idle power is 0.05 W, the task size is between 100 kb and 1000 kb, the bandwidth is 1 MHz versus 0.8 MHz, and the path loss index is 4. The energy consumption conversion factor is $\lambda_m^l = 1$ cents/J, $\lambda_m^n = 0.8$ cents/J. Finally, we set $\delta_0 = 1 \times 10^{-13}$, $N_0 = 3 \times 10^{-13}$.

- Random offloading scheme: randomly generates offloading decisions without considering the economic cost of users.
- Cost Optimization [19]: considers the economic cost of the users but does not consider the credibility of the MEC server or the urgency of the user task.
- AES scheme: Joint Optimization of resource allocation, dynamic pricing, and security offloading using AES cryptography.

Fig. 3 shows the distribution diagram of the simulated scenario of a

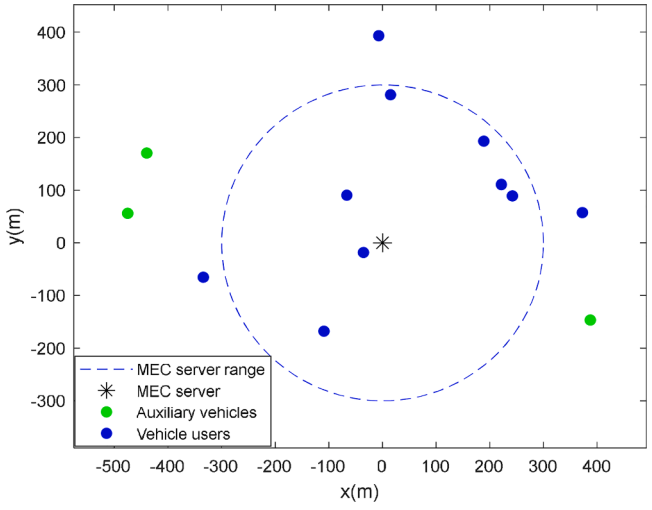


Fig. 3. Distribution of vehicles.

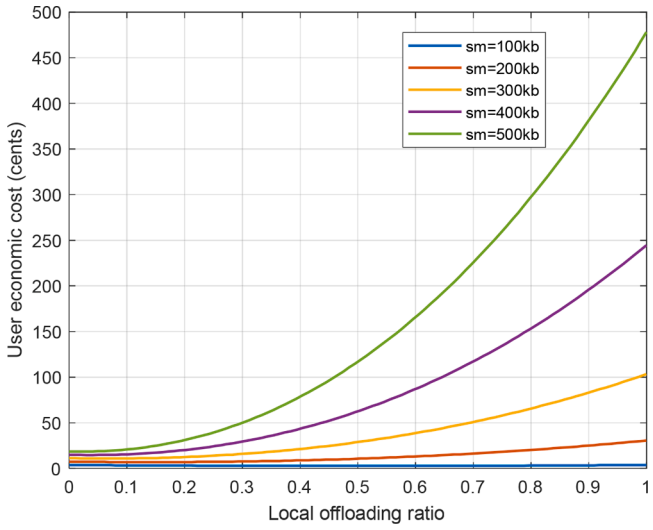


Fig. 4. Relationship between task volume and user economic costs.

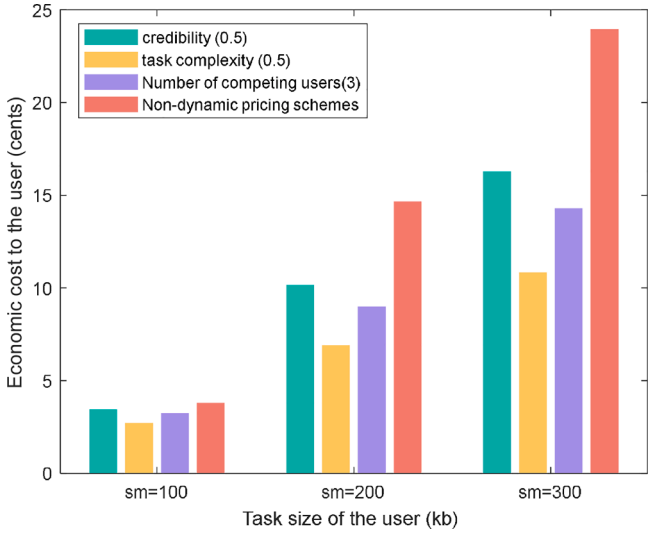


Fig. 5. Comparison of users economic costs of different scenarios.

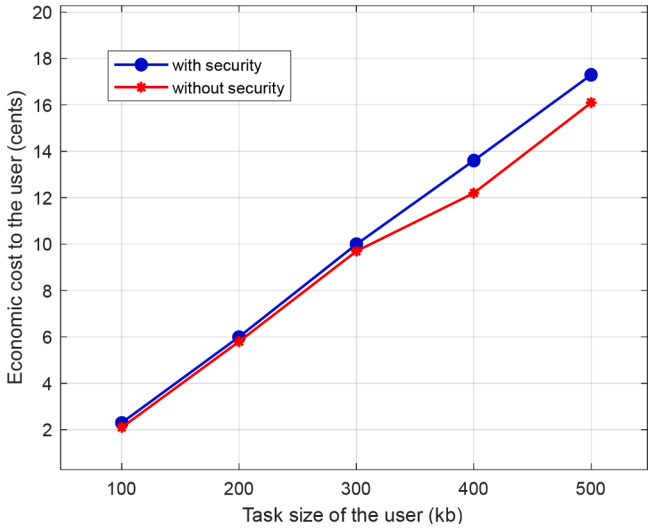


Fig. 6. Impact of user task data size on economic cost before and after homomorphic encryption for offloading tasks to auxiliary vehicles.

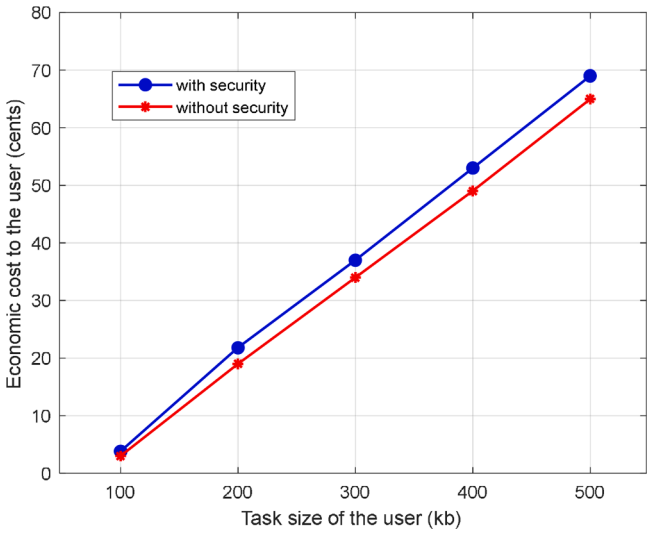


Fig. 7. Impact of user task data size on economic cost before and after homomorphic encryption for offloading tasks to MEC servers.

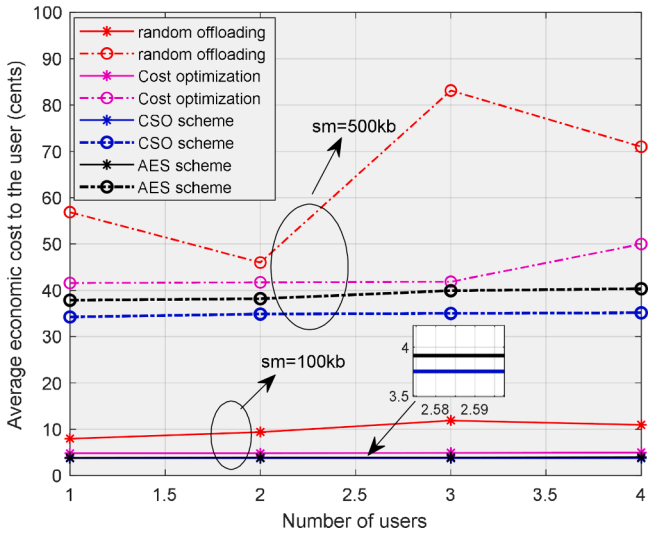


Fig. 9. Comparison of economic costs for users under different schemes.

single BS serving multiple users. The simulation scenario includes a BS equipped with a MEC server, 10 randomly distributed users, and 3 auxiliary vehicles with certain computational resources. As shown in Fig. 3, some users are within the coverage area of the MEC server and can offload tasks to the MEC server; some users are outside the range of the MEC server but there are auxiliary vehicles in their vicinity, and they can offload tasks to the auxiliary vehicles; some users are outside the range of the MEC server and there are no auxiliary vehicles in their vicinity, and thus they can only perform tasks locally. Fig. 3 is consistent with the presuppositions of this paper and with the reality.

Fig. 4 presents the relationship between the task volume of users and their economic costs when users select corresponding auxiliary vehicles to perform the offloading service. As shown in Fig. 4, it can be observed that an increase in the task volume of the user leads to an increase in the economic cost. This indicates that the user needs to weigh the relationship between task volume and economic cost when choosing the auxiliary vehicle to serve. As the task volume of the user increases, the user tends to process more tasks locally to minimize the economic cost; when the task volume is only 100 kb, the user prefers to process the task in the auxiliary vehicle.

Fig. 5 presents a comparison of the user economic cost of the CSO

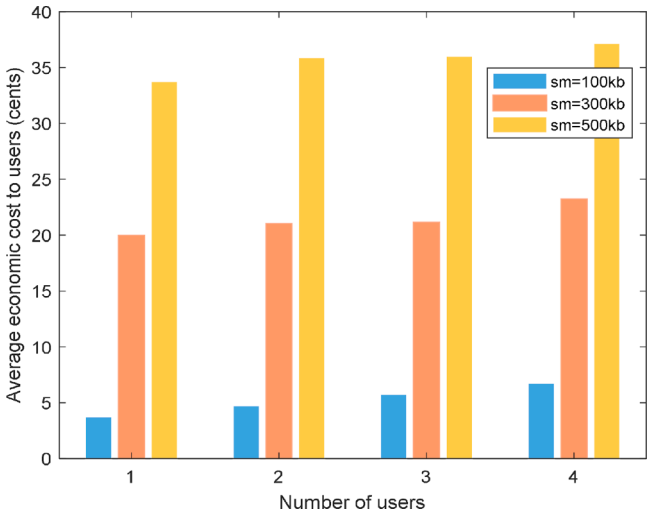


Fig. 8. Relationship between task size and economic cost for different number of users.

scheme with the traditional non-dynamic pricing scheme when the users select corresponding auxiliary vehicles for offloading service. For a better comparison, we plot the effects of the credibility of the auxiliary vehicle, the task complexity, and the number of users competing for auxiliary vehicles on the user economic cost, respectively. In our experiments, the credibility of the auxiliary vehicle is set to 0.5, the task complexity is set to 0.5, and the number of users competing for auxiliary vehicles is 3. From Fig. 5, we can observe that the user economic cost gradually increases as the task volume increases, and the user economic cost of the CSO scheme is much lower than that of the traditional pricing scheme, especially when the task volume is 300 kb. This indicates that the CSO scheme can effectively reduce the user's economic cost.

As shown in Figs. 6 and 7, it can be observed that as the number of tasks offloaded by the user increases, the economic cost of the user increases. The economic cost of the user is slightly higher when homomorphic encryption is used. This is because as the number of tasks increases, the energy consumption required for the encryption operation also increases. Therefore, the user needs to pay more to meet the energy requirements, which leads to an increase in the economic cost.

As shown in Fig. 8, when a user selects a MEC server for offloading services, we observe a trend that an increase in the number of users leads to a slight increase in the economic cost. Specifically, the economic cost of users gradually increases as the task volume increases. In our experiments, we set up 1 to 4 users, and the results show that the economic cost of users increases slightly as the number of users increases. This is because as the number of users increases, the resources of the MEC server that each user can obtain are relatively reduced. However, since the DDRP algorithm is effective, the increase in the economic cost of users is small.

Fig. 9 shows the comparison of the average economic cost of users under different schemes. In the experiments, we set the task sizes to 100 kb and 500 kb, respectively, and it is obvious from Fig. 9 that the CSO scheme significantly outperforms the cost-optimized scheme, the random offloading scheme, and the AES scheme. Specifically, when the task size is 100 kb, the CSO scheme reduces by 2.1 % compared to the Cost optimization scheme, and when the task size is 500 kb, the CSO scheme reduces by 12.3 % compared to the Cost optimization scheme. The reason why the CSO scheme outperforms the AES scheme in terms of economic cost is that the AES scheme involves both encryption at the user's sending end and decryption at the MEC server's receiving end, which increases the overall delay and energy consumption, and hence the economic cost. In contrast, the CSO scheme requires only encryption on the user side and no decryption on the receiving side of the MEC server, thus significantly reducing the economic cost. The simulation

results fully prove that the CSO scheme can effectively reduce the economic cost to the user. In addition, Fig. 9 shows that the advantage of the CSO scheme is more obvious when the task volume increases, which indicates that the winning DQN algorithm finds reasonable offloading and resource allocation decisions, further proving the effectiveness of the CSO scheme.

7. Conclusions

With the further improvement of ITS, billions of vehicles will continue to access the network in the future. MEC plays an increasingly important role in meeting the growing demand for computing resources from vehicles. To solve the problems of resource optimization and security offloading in vehicular edge computing, we propose for the first time the CSO scheme between auxiliary vehicles and MEC servers, which utilizes homomorphic encryption to protect user privacy and provide more comprehensive offloading services for users. In addition, we propose a dynamic computing resource pricing mechanism to ensure a fair transaction between service providers and users, considering the credibility of MEC servers and auxiliary vehicles, the urgency of the task, and the number of users competing for auxiliary vehicles. By optimizing the user selection decision, offloading decision, and resource allocation decision, we balance the resource utilization between auxiliary vehicles and MEC servers to achieve more efficient and secure computing services. Finally, in this paper, the DDRP algorithm and the CMSO algorithm are used to obtain the best offloading decision so that the economic cost of the user is minimized while securing the data transmission.

In the next step, we will conduct a deeper study of vehicle mobility models under OMA and NOMA systems in 6 G scenarios to ensure the efficiency and reliability of task offloading, and we will further study collaborative cache allocation strategy to optimize resource utilization. This research will lead to wider, safer, and more convenient applications of self-driving cars in the field of ITS.

CRediT authorship contribution statement

Jianbin Xue: Supervision. **Jia Yao:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Jiahao Wang:** Writing – review & editing, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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